H524 Introduction to Biostatistics

Week 1: Introduction to R, Data Presentation, and Summary Measures

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Outline

Course Overview

- **Course:** H524 Introduction to Biostatistics
- **Credits:** 4 credit hours
- **Schedule:**
  - Lecture: Monday, Wednesday 12:00-1:20pm
  - Lab: Wednesday 8:00am-9:50am
- **Textbook:** Principles of Biostatistics, 2nd edition (Pagano & Gauvreau)

What is Statistics?

Statistics provides methods and tools for collecting, analyzing, interpreting, and drawing inferences from sample data.

Two main branches:

- **Descriptive Statistics** – describe and summarize data
- **Inferential Statistics** – make inferences about populations from samples

Biostatistics: Application of statistics to biomedical and health data

- Broad interpretation of "Bio"
- Applied to drug testing, air pollution studies, nutrition, etc.
- *Example questions:* Is elevated air pollution associated with decreased lung function? Does a new treatment significantly lower risk compared to standard treatment?

Population vs. Sample

Population: The totality of subjects under study

- Often impossible or impractical to study entire population
- Contains the true *parameters* we want to learn about

Sample: The subset of the population actually studied

- Must be representative of the population
- Provides *statistics* that estimate population parameters

Key Concept: Statistical inference involves making assertions about population *parameters* based on sample *statistics*, subject to sampling error.

Learning Outcomes

By the end of this course, you will be able to:

1. Select and generate graphical and numerical summaries of data
2. Use principles of statistical inference to make conclusions about populations from samples
3. Communicate statistical findings to others
4. Use computer software to conduct statistical analysis

Evaluation

Component	**Weight**
Weekly Assignments	60%
Group Project	25%
Class Participation	15%

Why R for Biostatistics?

- **Free and Open Source:** No licensing costs
- **Comprehensive:** Extensive statistical capabilities
- **Reproducible Research:** Scripts can be saved and shared
- **Graphics:** Excellent data visualization capabilities
- **Community:** Large user base with extensive documentation
- **Packages:** Specialized biostatistics packages available

R Basics: Getting Started

- **Console:** Interactive command line
- **Scripts:** Save your work in .R files
- **Working Directory:** Know where your files are located
- **Help System:** Use `?function_name` or `help(function_name)`

Basic commands:

- `getwd()` - Check working directory
- `setwd()` - Set working directory
- `ls()` - List objects in workspace
- `rm()` - Remove objects

R Data Types

- **Numeric:** `x <- 3.14`
- **Integer:** `y <- 5L`
- **Character:** `name <- "John"`
- **Logical:** `flag <- TRUE`
- **Factor:** `gender <- factor(c("M", "F", "M"))`

Data structures:

- **Vector:** `c(1, 2, 3, 4)`
- **Matrix:** `matrix(1:6, nrow=2)`
- **Data Frame:** `data.frame(age=c(25,30), name=c("A","B"))`
- **List:** `list(numbers=1:5, text="hello")`

Data Sources and Citations

Real datasets from published sources:

- **FEV1 (Lung Function) Data**
  - Source: Pagano & Gauvreau "Principles of Biostatistics", pp. 38-39
  - Study of 13 adolescents with asthma
  - Variables: FEV1 measurements (liters), gender
- **Childhood Injury Deaths**
  - Source: Pagano & Gauvreau "Principles of Biostatistics", pp. 24-25
  - 100 children aged 5-9 who died from injuries, 1980-1985
  - Variables: Cause of death (categorical)

Additional sources available:

- Serum cholesterol: National Health Examination Survey (pp. 12-15)
- Birth weight: 3.7 million US births, 1986 (pp. 26-28)
- All data publicly available through government agencies and published studies

Types of Data: Quantitative

- **Quantitative (Numerical Data)**
  - *Continuous:* Can take any value within a range
    - Blood pressure (120.5 mmHg)
    - Body temperature (98.6°F)
    - Cholesterol level (185.3 mg/dL)
    - Time to recovery (3.5 days)
  - *Discrete:* Countable values
    - Number of hospitalizations (0, 1, 2, 3...)
    - Pill count in a bottle (30, 60, 90...)
    - Number of symptoms reported (1, 2, 3...)
    - Children in family (0, 1, 2, 3...)

Types of Data: Qualitative

- **Qualitative (Categorical Data)**
  - *Nominal:* No natural ordering
    - Blood type (A, B, AB, O)
    - Gender (Male, Female, Other)
    - Insurance type (Private, Medicare, Medicaid)
    - Medication name (Aspirin, Ibuprofen)
  - *Ordinal:* Natural ordering exists
    - Pain scale (None, Mild, Moderate, Severe)
    - Disease stage (Stage I, II, III, IV)
    - Education level (Less than HS, HS, College, Graduate)
    - Smoking status (Never, Former, Current)

Frequency Distributions

Purpose: Organize and summarize data

For Categorical Data:

- Frequency tables
- Relative frequency (proportions)
- Percentages

For Continuous Data:

- Group data into intervals (bins)
- Create frequency tables
- Choose appropriate number of intervals

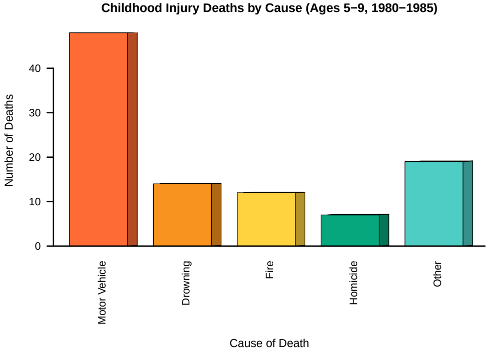
Graphical Displays: Categorical Data

- **Bar Charts**
  - Bars separated by spaces
  - Height represents frequency or percentage
  - Good for comparing categories
- **Pie Charts**
  - Shows proportions of a whole
  - Best for few categories
  - Less effective for precise comparisons

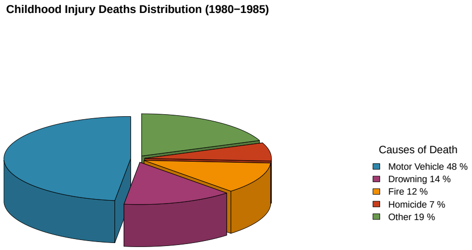
R Example:

```
# Create bar chart for education levels
education_counts <- table(survey_data$education)
barplot(education_counts, main = "Education Distribution",
        ylab = "Frequency", col = c("#FF6B35", "#F7931E"))

# Create pie chart for insurance types
insurance_counts <- table(survey_data$insurance)
pie(insurance_counts, main = "Insurance Distribution")
```

Bar Chart Example



Pie Chart Example

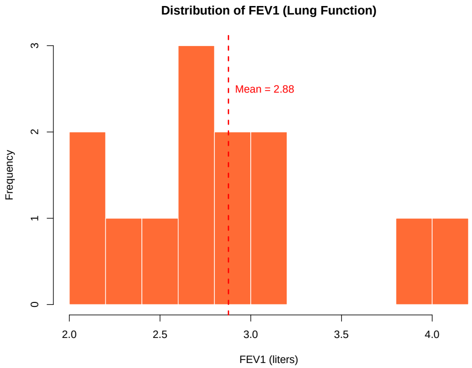


- **Histograms**
  - Bars touch each other
  - Shows distribution shape
  - Data is grouped into bins (intervals)
  - Bin width affects appearance and is somewhat subjective
  - Different bin choices can reveal different patterns
- **Box Plots**
  - Shows median, quartiles, outliers
  - Good for comparing groups
  - Compact display
- **Dot Plots**
  - Each observation as a dot
  - Good for small datasets

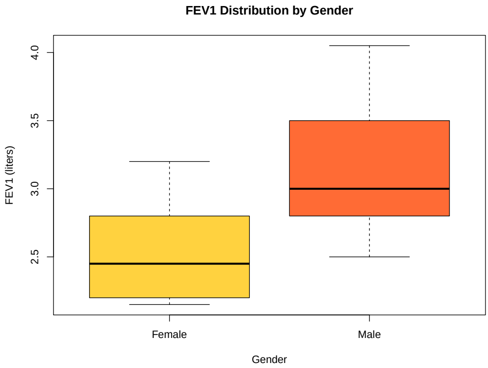
R Example:

```
# Histogram for BMI distribution
hist(health_data$bmi, main = "BMI Distribution",
     xlab = "BMI", col = "#FF6B35")

# Box plot for blood pressure
boxplot(health_data$systolic_bp, main = "BP Distribution")
```

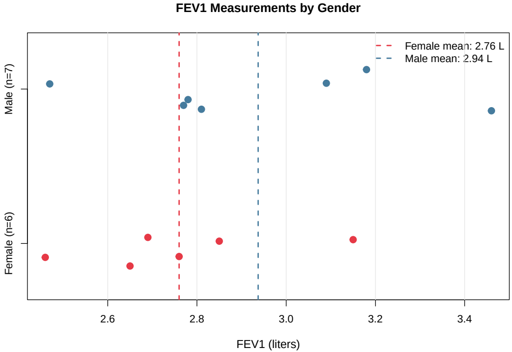
Histogram Example



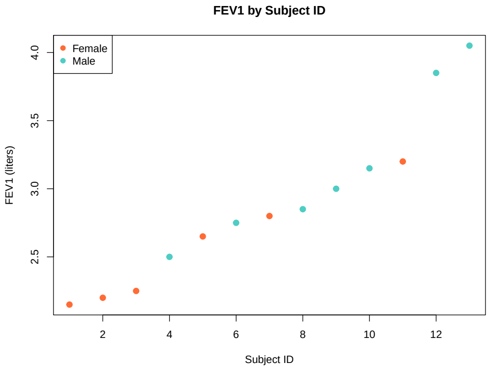
Box Plot Example



Dot Plot Example



Scatter Plot Example



Measures of Central Tendency

- **Mean ($\bar{x}$)**
  - $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$
  - Sensitive to outliers
  - Most common measure
- **Median**
  - Middle value when data is ordered
  - Resistant to outliers
  - Good for skewed distributions
- **Mode**
  - Most frequently occurring value
  - Can have multiple modes
  - Useful for categorical data

Measures of Variability

- **Range**
  - Maximum - Minimum
  - Simple but sensitive to outliers
- **Variance (s^2)**
  - $s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$
  - Average squared deviation from mean
- **Standard Deviation (s)**
  - $s = \sqrt{s^2}$
  - Same units as original data

Empirical Rule (68-95-99.7 Rule)

When data are symmetric and unimodal (bell-shaped):

- Approximately **68%** of observations fall within **1 standard deviation** of the mean
- Approximately **95%** fall within **2 standard deviations** of the mean
- Approximately **99.7%** fall within **3 standard deviations** of the mean

Application: Helps identify unusual observations and understand data distribution patterns

Example: If blood pressure has mean = 120 mmHg and SD = 15 mmHg:

- 68% of people have BP between 105-135 mmHg
- 95% have BP between 90-150 mmHg

Percentiles and Quartiles

- **Percentiles:** Value below which a certain percentage of data falls
- **Quartiles:** Divide data into four equal parts
  - Q1 (25th percentile): First quartile
  - Q2 (50th percentile): Median
  - Q3 (75th percentile): Third quartile
- **Interquartile Range (IQR):** Q3 - Q1
  - Contains middle 50% of data
  - Resistant to outliers
- **Five-Number Summary:** Min, Q1, Median, Q3, Max

Data Collection and Sampling

Representative Sample: Distribution of values in sample mimics population distribution

Common Sampling Methods:

- **Simple Random Sample (SRS):** Every group of size n has same chance of selection
- **Stratified Random Sample:** Divide population into strata, then SRS from each stratum
- **Cluster Sampling:** Divide into clusters, select random clusters, study everyone in selected clusters
- **Sample of Convenience:** Easy to obtain but may not be representative

Key Point: Non-representative samples can be very misleading due to sampling bias

Introduction to Rates

Rate: A measure of the frequency of occurrence of an event

General form:

$$\text{Rate} = \frac{\text{Number of events}}{\text{Population at risk}} \times k$$

where k is a constant (often 1,000, 10,000, or 100,000)

Why use rates?

- Enable comparison between populations of different sizes
- Standardize measurements across time periods
- Account for population at risk

Common Types of Rates

- **Crude Birth Rate**

$$\frac{\text{Number of births in a year}}{\text{Total population}} \times 1000$$

- **Crude Death Rate**

$$\frac{\text{Number of deaths in a year}}{\text{Total population}} \times 1000$$

- **Disease Incidence Rate**

$$\frac{\text{New cases of disease}}{\text{Population at risk}} \times k$$

- **Case Fatality Rate**

$$\frac{\text{Deaths from disease}}{\text{Total cases of disease}} \times 100$$

Age-Specific Rates

Problem: Crude rates can be misleading when comparing populations with different age structures

Solution: Calculate rates for specific age groups

$$\text{Age-specific rate} = \frac{\text{Events in age group}}{\text{Population in age group}} \times k$$

Advantages:

- More precise comparisons
- Accounts for age distribution differences
- Reveals age-related patterns

Age Standardization

Direct Method:

1. Apply age-specific rates from each population to a standard population
2. Sum across age groups
3. Compare standardized rates

$$\text{Directly standardized rate} = \frac{\sum_i R_i \times P_i^{std}}{\sum_i P_i^{std}} \times k$$

where:

- R_i = age-specific rate for age group i
- P_i^{std} = standard population size for age group i

Age Standardization (continued)

Indirect Method:

1. Apply standard age-specific rates to study population
2. Calculate expected number of events
3. Compare observed vs. expected

$$\text{SMR} = \frac{\text{Observed events}}{\text{Expected events}}$$

SMR = Standardized Mortality Ratio

- SMR > 1: Higher than expected
- SMR < 1: Lower than expected
- SMR $= 1$: Same as expected

AI in Biostatistics: A Modern Learning Tool

- **AI tools can enhance learning** but cannot replace understanding
- **Use AI as a statistical tutor** that requires supervision
- **Critical evaluation** of AI output is essential
- **Maintain statistical rigor** while leveraging AI assistance

Today's Goal: Learn to use AI effectively for biostatistics

Popular AI Tools:

- ChatGPT, Claude (used in today's demos), Copilot
- R-specific: GitHub Copilot for R
- Statistical help: Wolfram Alpha

AI Demo 1: Generating R Code

Prompt: "Create R code to calculate descriptive statistics for blood pressure data"

Claude AI Response:

```
# First, load the real data from our Week 1 script
source("week1_lecture_code.R")
# This creates fev1_data with FEV1 measurements from real study

# Calculate descriptive statistics for FEV1 data
mean_fev1 <- mean(fev1_data$fev1_liters, na.rm = TRUE)
median_fev1 <- median(fev1_data$fev1_liters, na.rm = TRUE)
sd_fev1 <- sd(fev1_data$fev1_liters, na.rm = TRUE)
min_fev1 <- min(fev1_data$fev1_liters, na.rm = TRUE)
max_fev1 <- max(fev1_data$fev1_liters, na.rm = TRUE)

# Display results
cat("Mean FEV1:", round(mean_fev1, 2), "liters\n")
cat("Median FEV1:", round(median_fev1, 2), "liters\n")
cat("SD:", round(sd_fev1, 2), "liters\n")
cat("Range:", min_fev1, "-", max_fev1, "liters\n")

# Alternative: use summary function
summary(fev1_data$fev1_liters)
```

What to check: Variable names, missing value handling, appropriate functions

AI Demo 2: Creating Visualizations

Prompt: "Create a professional histogram of blood pressure data with normal curve overlay"

Claude AI Response:

```
# Assumes bp_data is already loaded (from week1_lecture_code.R)
# Create histogram with normal curve overlay
hist(bp_data$systolic_bp,
     breaks = 15,
     probability = TRUE,  # Use density instead of frequency
     main = "Distribution of Systolic Blood Pressure",
     xlab = "Systolic BP (mmHg)",
     ylab = "Density",
     col = "lightblue",
     border = "white")

# Add normal curve overlay
x_vals <- seq(min(bp_data$systolic_bp), max(bp_data$systolic_bp), length = 100)
mean_val <- mean(bp_data$systolic_bp)
sd_val <- sd(bp_data$systolic_bp)
y_vals <- dnorm(x_vals, mean = mean_val, sd = sd_val)
lines(x_vals, y_vals, col = "red", lwd = 2)

# Add vertical line for mean
abline(v = mean_val, col = "darkred", lwd = 2, lty = 2)
legend("topright", c("Normal Curve", "Mean"),
       col = c("red", "darkred"), lwd = 2, lty = c(1, 2))
```

AI Demo 3: Explaining Statistical Concepts

Student Question: "When should I use median instead of mean for blood pressure data?"

Claude AI Response:

- **Use median when:**
  - Data has outliers (very high/low BP readings)
  - Distribution is skewed (not bell-shaped)
  - Want robust measure unaffected by extreme values
- **Use mean when:**
  - Data is approximately normally distributed
  - No significant outliers present
  - Need measure that uses all data points
- **For blood pressure:** Often use median because BP can have extreme values due to measurement errors or acute conditions

Always verify: Check your actual data distribution before deciding!

AI Demo 4: Interpreting Results

Prompt: "Interpret these blood pressure analysis results for a health class"

```
Summary Statistics:
Mean: 133.6 mmHg, SD: 18.6 mmHg
Median: 131.0 mmHg, Range: 96-196 mmHg
Males: Mean = 136.2 mmHg, Females: Mean = 130.8 mmHg
```

Claude AI Interpretation:

- **Central tendency:** Average BP (133.6) is in "high normal" range (AHA guidelines: ¡120 normal, 120-129 elevated, 130+ high)
- **Variability:** Large SD (18.6) indicates substantial individual differences
- **Distribution:** Mean ¿ median suggests slight right skew (some high outliers)
- **Gender difference:** Males average 5.4 mmHg higher (typical pattern)
- **Range:** Wide range (96-196) includes both hypotensive and severely hypertensive readings

Clinical relevance: About 30% of this population likely has hypertension requiring medical attention

AI Best Practices for Biostatistics

DO:

- Start with specific, clear prompts
- Ask for explanations of code
- Request multiple approaches
- Verify results independently
- Ask for biological context
- Use AI to check your work

DON'T:

- Blindly copy-paste code
- Skip understanding the logic
- Use without domain knowledge
- Ignore warning messages
- Accept first response
- Replace critical thinking

Golden Rule: AI should enhance your learning, not replace it

Common AI Pitfalls in Statistics

- **Outdated Information:** AI training data may not include latest statistical methods
- **Context Confusion:** May suggest inappropriate tests for biomedical data
- **Assumption Violations:** Doesn't always check statistical assumptions
- **Missing Values:** May not handle missing data appropriately
- **Interpretation Errors:** Can confuse correlation with causation
- **Package Dependencies:** May suggest functions from packages you don't have

Always:

- Check your data fits the suggested method
- Verify assumptions are met
- Cross-reference with textbooks/course materials
- Test code on small examples first

Effective Prompting for Biostatistics

Poor Prompt: `‘‘Help with statistics’’`

Better Prompt: `‘‘I have blood pressure data from 100 patients. Create R code to compare BP between males and females using appropriate statistical test.’’`

Best Prompt:

```
‘‘I have blood pressure data from 100 patients with variables:
age, gender, systolic_bp, treatment_group. I want to:
1. Compare systolic BP between males and females
2. Check if data meets normality assumptions
3. Choose appropriate statistical test
4. Interpret results in clinical context
5. Create publication-quality plot
Please provide R code with explanations.’’
```

Key elements: Context, sample size, variables, specific goals, level of explanation needed

AI Assignment Preview

This week's homework will include:

1. Using AI to generate R code for data analysis
2. Comparing AI output with manual coding
3. Identifying and correcting AI errors
4. Writing effective statistical prompts
5. Critical evaluation of AI explanations

Learning Goals:

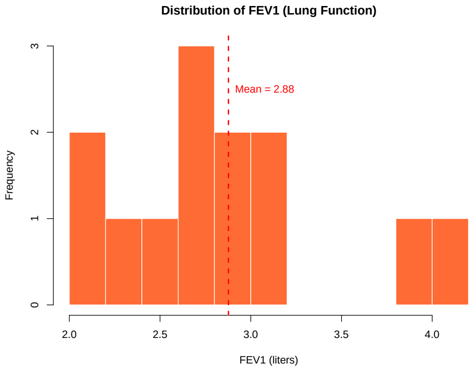
- Develop AI literacy for biostatistics
- Maintain statistical rigor while using AI tools
- Build confidence in statistical computing
- Prepare for modern data science workflows

Remember: AI is a powerful tool, but you are the biostatistician!

R Example: Blood Pressure Study Data

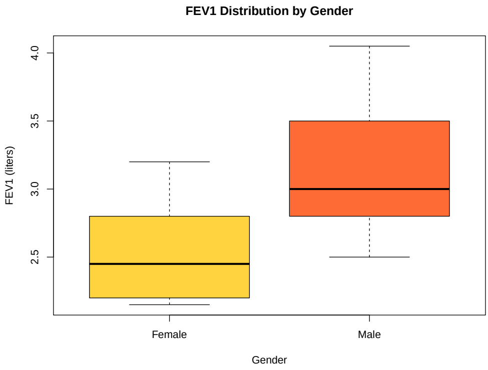
```
# To follow along: source the complete script
source("week1_lecture_code.R")

# Or generate the data yourself:
set.seed(123)
bp_data <- data.frame(
  age = round(rnorm(100, mean = 45, sd = 15)),
  gender = sample(c("Male", "Female"), 100, replace = TRUE),
  systolic_bp = round(rnorm(100, mean = 130, sd = 20)),
  treatment = sample(c("Control", "Treatment A", "Treatment B"),
                     100, replace = TRUE)
)

# Basic descriptive statistics
mean(bp_data$systolic_bp)     # Mean: 133.6 mmHg
sd(bp_data$systolic_bp)       # SD: 18.6 mmHg
summary(bp_data$systolic_bp)  # Five-number summary
```

Data Visualization: Histogram



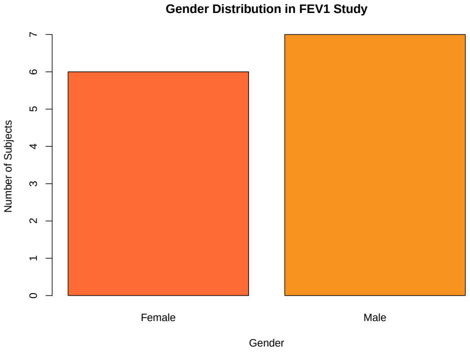
- Shows distribution of FEV1 lung function measurements
- Red dashed line indicates mean value (2.88 liters)
- Real data from 13 adolescents with asthma

Data Visualization: Box Plots



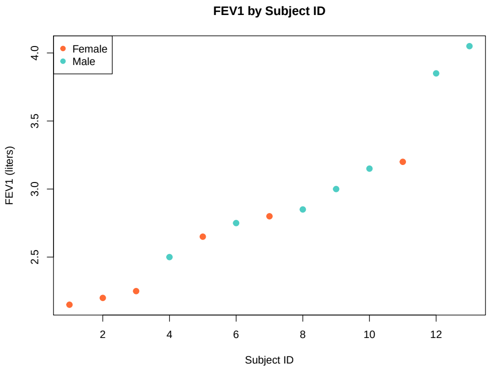
- Compares FEV1 distributions between genders
- Shows median, quartiles, and outliers
- Males have higher median FEV1 than females

Data Visualization: Bar Charts



- Good for categorical data
- Shows frequency counts for gender in FEV1 study
- 6 females, 7 males in the study

Data Visualization: Scatter Plots



- Shows FEV1 by subject ID, colored by gender
- No clear pattern with subject ID
- Illustrates gender differences in lung function

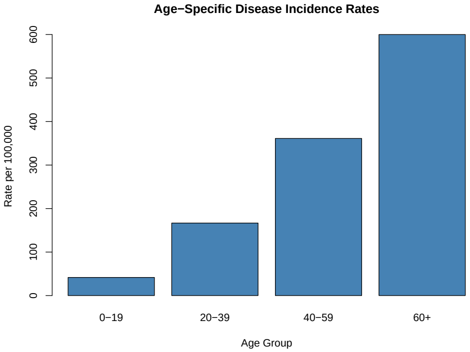
R Code: Creating Visualizations

```r
# Histogram
hist(bp_data$systolic_bp,
     main = "Distribution of Systolic Blood Pressure",
     xlab = "Systolic BP (mmHg)", col = "lightblue")

# Box plot by gender
boxplot(systolic_bp ~ gender, data = bp_data,
        main = "Systolic Blood Pressure by Gender",
        col = c("pink", "lightblue"))

# Scatter plot with regression line
plot(bp_data$age, bp_data$systolic_bp,
     xlab = "Age (years)", ylab = "Systolic BP (mmHg)")
abline(lm(systolic_bp ~ age, data = bp_data), col = "red")
```

R Example: Rates Calculation

```
# Disease surveillance data
population_size <- 50000
new_cases <- 125
deaths <- 15

# Calculate incidence rate
incidence_rate <- (new_cases / population_size) * 100000
# Result: 250 per 100,000 population

# Calculate case fatality rate
case_fatality_rate <- (deaths / new_cases) * 100
# Result: 12%

# Age-specific rates
population_by_age <- c(12000, 15000, 18000, 5000)
cases_by_age <- c(5, 25, 65, 30)
age_specific_rates <- (cases_by_age / population_by_age) * 100000
```

Age-Specific Rates Visualization



- Clear age gradient in disease incidence
- Highest rates in oldest age group (600 per 100,000)
- Demonstrates importance of age-specific analysis

Summary

Today we covered:

- Introduction to R and its capabilities
- Types of data and appropriate presentation methods
- Graphical displays for different data types
- Numerical summary measures (central tendency and variability)
- Rates and age standardization methods

Next week: We'll dive deeper into probability concepts

Reading: Pagano & Gauvreau, Chapters 1-4

Questions?

Thank you!

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